



E4 Educator v2.0

Project Specification

Walark Electronics · Fundrum Engineering

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Board revision: E4-EDU-v101 → E4-EDU-v200

1 Board overview

The E4 Educator v2.0 is a purpose-designed ESP32 teaching platform produced by Walark Electronics under the Fundrum Engineering curriculum framework. It is the second revision of the Enceladus Educator board series, incorporating lessons learned from v1.0 field use and structured to support a progressive three-tier course from PlatformIO fundamentals through to integrated firmware projects.

The board is designed around the principle that every component present must appear in at least one tutorial. Nothing is fitted without a teaching purpose.

1.1 Design philosophy

- Spec-first discipline — board design drives the curriculum, not the reverse
- Socketed architecture — ESP32 module, TFT display, and OLED are all removable and swappable
- Progressive disclosure — OLED introduces I2C and simple display; TFT introduces SPI, touch, and SD card as a later milestone
- No unexplained silkscreen — every label, pin, and connector appears in a tutorial
- Fundrum conventions throughout — pinned platform, `build_flags` for all GPIO, `millis()` timing, semantic versioning

1.2 Key specifications

Parameter	Detail
Processor	ESP32 DevKit V1 — dual core Xtensa LX6, 240MHz, 4MB flash
Framework	Arduino on ESP32 via PlatformIO, espressif32@6.6.0 pinned
Display (primary)	2.8" ILI9341 TFT SPI 240×320 with XPT2046 touch and microSD slot, socketed
Display (entry)	0.96" SSD1306 OLED 128×64, I2C, socketed — course entry point
Onboard sensors	DS18B20 one-wire, AHT20 I2C temp/humidity, MAX4466 analog mic, ICS43434 MEMS I2S mic, LDR light sensor
I2C expansion	Two expansion headers (AHT20-BME280-BH1750 and std I2C), plus QWIIC-compatible pinout
Audio output	DAC1 GPIO 25, piezo buzzer GPIO 27, PWM transistor output GPIO 12
User controls	NEXT button GPIO 13, ENT button GPIO 14, BOOT and RST duplicated on front face
LEDs	RED GPIO 16, GREEN GPIO 26, BLUE GPIO 32
Connectivity	WiFi 802.11 b/g/n, Bluetooth 4.2 BLE + Classic (ESP32 onboard)
Power	USB-C 5V input, 3.3V regulation onboard. 3V3 devices only.
Form factor	Single PCB, ESP32 rear-mounted in SIL sockets, TFT upper half, controls bottom edge

2 Changes from v1.0

The following changes have been made from the E4-EDU-v101 board to produce the v2.0 revision.

Type	Item	Change	Reason
Fix	RED LED	GPIO 3 → GPIO 16	GPIO 3 is UART0 RX — conflicts with Serial monitor
Fix	GREEN LED	GPIO 1 → GPIO 26	GPIO 1 is UART0 TX — conflicts with Serial.print()
Fix	BLUE LED	GPIO 15 → GPIO 32	GPIO 15 boot sensitive. GPIO 32 is clean ADC1 pin
New	OLED socket	4-pin I2C header for 0.96" SSD1306	Entry-level display. I2C bus, no new GPIO pins required
New	BOOT + RST	Duplicate tactile buttons on front face	Avoids flipping board during frequent reflash cycles
New	SD_CS	GPIO 15 assigned to TFT module microSD CS	Freed by LED reassignment. 10k pull-up to 3V3 required
New	UART header	3-pin TX/RX/GND on freed GPIO 1 and 3	Enables serial comms tutorials and logic analyser use
Revised	Test points	TP1/TP2/TP4 removed. Remaining 8 moved to right edge	Frees space under TFT for OLED and BOOT/RST
Revised	TFT connector	Fixed header → socketed female connector	TFT removable for OLED-only tutorials
Revised	ESP32 mount	Rear-mounted in turned-pin SIL sockets	Module swappable between projects without reflashing

3 GPIO assignment table

All GPIO assignments for the E4 Educator v2.0. Pins marked Input Only cannot drive outputs — they are handed to hardware peripherals (I2S, ADC) which bypass this restriction. GPIO 6–11 are reserved for internal flash and must never be used.

GPIO	Assignment	Status	Notes
GPIO 16	RED LED	Clean	Verify: <code>ESP.getPsramSize() == 0</code> (no PSRAM on DevKit V1)
GPIO 26	GREEN LED	Clean	DAC2 capable. No conflicts.
GPIO 32	BLUE LED	Shared	Shared with I2S BCLK. Do not use MEMS mic and BLUE LED simultaneously.
GPIO 13	NEXT button	Clean	INPUT_PULLUP. Front face.
GPIO 14	ENT button	Clean	INPUT_PULLUP. Front face.
GPIO 0	BOOT (front duplicate)	Caution	Boot sensitive. Parallel to module BOOT. Pull-up only, never drive LOW at boot.
EN	RST (front duplicate)	Clean	EN net, not a GPIO. Safe to expose as button.
GPIO 23	SPI MOSI	Clean	Shared: TFT, Touch, SD card
GPIO 18	SPI SCK	Clean	Shared: TFT, Touch, SD card
GPIO 19	SPI MISO	Clean	Shared: TFT, Touch, SD card
GPIO 5	TFT_CS	Caution	Boot sensitive — PWM at power-up. TFT_eSPI handles correctly.
GPIO 17	TFT_DC	Clean	
GPIO 33	TOUCH_CS	Clean	XPT2046 touch controller. ADC1 Ch5.
GPIO 15	SD_CS	Caution	Boot sensitive. Add 10k pull-up to 3V3 if not on TFT module. Init SD after TFT.
GPIO 21	I2C SDA	Clean	AHT20, BMP280, OLED, I2C expansion headers
GPIO 22	I2C SCL	Clean	AHT20, BMP280, OLED, I2C expansion headers
GPIO 4	DS18B20 one-wire	Clean	4.7k pull-up fitted on board
GPIO 36	MAX4466 analog mic	Input only	ADC1 VP
GPIO 39	LDR light sensor	Input only	ADC1 VN
GPIO 32	I2S BCLK (MEMS)	Shared	Shared with BLUE LED — see above
GPIO 35	I2S WS (MEMS)	Input only	I2S peripheral bypasses input-only restriction. Works correctly.
GPIO 34	I2S SD (MEMS)	Input only	I2S data input
GPIO 25	DAC1 / PWM audio	Clean	True analog output. DAC1.
GPIO 27	Piezo buzzer	Clean	PWM tone via LEDC
GPIO 12	PWM transistor (BC847)	Caution	Bootstrap pin. Keep LOW at boot. Safe in operation at 3V3.

GPIO 1	UART0 TX / UART header	Freed	Was GREEN LED in v1. Now UART debug header TX pin.
GPIO 3	UART0 RX / UART header	Freed	Was RED LED in v1. Now UART debug header RX pin.

Note: GPIO 6–11 are hard-wired to internal flash SPI and must never be used in any project.

4 platformio.ini build flags

All E4 Educator v2.0 projects must use the following `build_flags` block. GPIO numbers are never hardcoded in firmware — always reference these defines. The platform version is pinned to prevent silent breaking changes from upstream updates.

4.1 platformio.ini template

```
[env:e4_educator_v2]
platform      = espressif32@6.6.0
board         = esp32dev
framework    = arduino
monitor_speed = 115200
monitor_port  = COM10          ; adjust per machine
upload_port   = COM10          ; adjust per machine

build_flags =
    -DFW_VERSION=\"1.0.0\"
    -DFW_DATE=\"2026-05-01\"
    ; LEDs
    -DLED_RED=16
    -DLED_GREEN=26
    -DLED_BLUE=32
    ; Buttons
    -DBTN_NEXT=13
    -DBTN_ENT=14
    ; TFT SPI display
    -DTFT_CS=5
    -DTFT_DC=17
    -DTFT_MOSI=23
    -DTFT_SCLK=18
    -DTFT_MISO=19
    ; Touch and SD card
    -D TOUCH_CS=33
    -DSD_CS=15
    ; I2C bus
    -DI2C_SDA=21
    -DI2C_SCL=22
    ; Sensors
    -DONE_WIRE=4
    -DLDR=39
    -DMIC_ANALOG=36
    ; MEMS I2S microphone
    -DMEMS_BCLK=32
    -DMEMS_WS=35
    -DMEMS_SD=34
    ; Audio output
    -DDAC_AUDIO=25
    -DPIEZO=27
```

Note: Copy monitor_port and upload_port to a local platformio.ini override if working across multiple machines. Never commit machine-specific paths to version control.

5 Course curriculum outline

The E4 Educator curriculum is structured in three tiers. Each tier builds directly on the previous. Tier 1 establishes the toolchain and board. Tier 2 develops the core skills used in every real project. Tier 3 combines those skills into complete, inspiring integrated projects.

5.1 Tier 1 — The platform

Establishes PlatformIO, the E4 board, and the Fundrum development conventions. Target audience: experienced software developers or electronics hobbyists new to embedded systems.

T	Title	Skills covered
T01	PlatformIO setup and first build	platformio.ini, espressif32@6.6.0 pin, USB driver, serial monitor
T02	GPIO and build_flags	LED blink via defines, millis() non-blocking timing, no delay()
T03	Buttons and debouncing	INPUT_PULLUP, debounce logic, short/long press, state machines
T04	I2C fundamentals	I2C scanner, bus addressing, AHT20 temp/humidity, Wire library
T05	OLED display	SSD1306, u8g2/Adafruit library, layout with 128×64 constraints
T06	One-wire and analog	DS18B20 one-wire, LDR ADC1 read, ADC1 vs ADC2 WiFi rule
T07	Audio output basics	Piezo tone via LEDC, DAC sine wave, PWM transistor load control

5.2 Tier 2 — Core skills

Develops the building blocks used in every real firmware project. These tutorials are self-contained but cross-reference each other.

T	Title	Skills covered
T08	TFT display and sprite pipeline	TFT_eSPI, User_Setup.h, sprite double-buffer, zones, smooth fonts
T09	Touch input	XPT2046, calibration, touch zones, SPI bus sharing with TFT
T10	SD card and file I/O	SD library, SPI init order (TFT first), read/write, BMP splash screen
T11	NVS persistence	Preferences library, NVS namespace, key ≤15 chars, reboot survival
T12	WiFi and MQTT	WiFi connect, PubSubClient, publish/subscribe, MarkPi3 broker
T13	LittleFS	Partition config, file upload tool, serve web page, store config
T14	OTA updates	ElegantOTA, AsyncWebServer, update workflow, version display

T15	ESP-NOW	Peer-to-peer without infrastructure, Fundrum ESP-NOW protocol v1.0.0
T16	I2S audio and MEMS mic	ICS43434 MEMS mic, I2S peripheral, FFT basics, GPIO 32/34/35

5.3 Tier 3 — Integrated projects

Complete, inspiring projects that combine Tier 1 and Tier 2 skills. Each project produces a finished, usable piece of firmware.

P	Project	Skills combined
P01	Environmental monitor	AHT20 + DS18B20, TFT dashboard, MQTT to broker, OTA, NVS calibration
P02	Data logger	Multi-sensor, SD card CSV logging, RTC timestamp, LittleFS web dashboard
P03	Wireless sensor node	ESP-NOW star network, multiple E4 nodes, master TFT display, NVS provisioning
P04	Audio spectrum analyser	MEMS mic, I2S, FFT, TFT real-time bar graph display, MQTT peak data
P05	Touch UI controller	Multi-screen TFT app, touch navigation, NVS settings, OTA, MQTT control

6 Socketed architecture

A key pedagogical feature of the E4 Educator v2.0 is that the three principal components — ESP32 module, TFT display, and OLED display — are all socketed and removable. This solves a genuine learner problem.

6.1 ESP32 module

The ESP32 DevKit V1 module is rear-mounted in turned-pin SIL sockets. A learner can pop the module out and keep it in a labelled anti-static bag with a working project on it, then drop a fresh module in and continue learning. This teaches the correct mental model — firmware lives on the chip, not the board.

Note: Verify GPIO 16 is free by running `ESP.getPsramSize()` — must return 0. WROVER variants use GPIO 16/17 for PSRAM and are not compatible with this board revision.

6.2 Display progression

The course deliberately introduces the OLED before the TFT. The OLED connects via I2C (two wires, no configuration file) and its 128×64 resolution forces layout discipline. When the TFT is introduced in Tier 2 it feels like a reward — colour, sprites, touch, and SD card access. The contrast makes the TFT more impressive and the learning curve feels earned.

	0.96" OLED (Tier 1)	2.8" TFT (Tier 2)
Interface	I2C — SDA/SCL only	SPI — MOSI/SCK/MISO/DC/CS
Configuration	Library address (0x3C)	User_Setup.h with pin defines
Resolution	128×64 monochrome	240×320 16-bit colour
Additional	None	XPT2046 touch + microSD slot

Course tier	Tier 1 T05	Tier 2 T08–T10
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7 Version information

Version	Date	Author	Changes
1.0	May 2026	Mark Weir	Initial specification for E4-EDU-v200 revision

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